UNIQUENESS IN THE VALUATIVE CRITERION IS EQUIVALENT TO SEPARATEDNESS OF THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Allowed standard prerequisites (editorial).

- The description of the diagonal as a morphism to which the valuative criterion applies. Used throughout the counted proof (source lines 9595-9608). Reduces the uniqueness statement for the morphism to a statement about its diagonal.
- The translation of morphisms of dotted arrows for the diagonal into pairs of dotted arrows for the morphism. Used inside the counted proof (source lines 9609-9618). Identifies the two formulations of uniqueness.

Source reference. Original definition definition-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 601–601. Navigation pointer only: the counted text refers to the Stacks item definition-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-helper-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9543–9543. Navigation pointer only: the counted text refers to the Stacks item lemma-helper-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-properties-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 344–344. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The

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Source reference. Original lemma lemma-setoids-and-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9451–9451. Navigation pointer only: the counted text refers to the Stacks item lemma-setoids-and-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-quasi-compact-existence-universally-closed: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9527–9527. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-quasi-compact-existence-universally-closed, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume f is quasi-separated. If f satisfies the uniqueness part of the valuative criterion, then f is separated.*

Proof. The assumption on f means that Δ_f is quasi-compact and quasi-separated (Definition definition-separated). We have to show that Δ_f is proper. Lemma lemma-setoids-and-diagonal says that Δ_f is separated. By Lemma lemma-properties-diagonal we know that Δ_f is locally of finite type. To finish the proof we have to show that Δ_f is universally closed. A formal argument (see Lemma lemma-helper-diagonal) shows that the uniqueness part of the valuative criterion implies that we have the existence of a dotted arrow in any solid diagram like so:

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathcal{X} \\ \downarrow & \nearrow & \downarrow{\scriptstyle \Delta_f} \\ \mathrm{Spec}(A) & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

Using that this property is preserved by any base change we conclude that any base change by Δ_f by an algebraic space mapping into $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ has the existence part of the valuative criterion and we conclude is universally closed by the valuative criterion for morphisms of algebraic spaces, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-existence-universally-closed. □